

Data Communication

Chapter-02

The OSI Model

The **OSI model** (Open Systems Interconnection model) is an important standard created by the International Organization for Standardization (ISO). It helps us understand how different computer systems and devices communicate over a network.

The OSI model divides this communication process into **seven layers**, and each layer has its own specific job. This makes it easier to design networks and troubleshoot problems when they arise.

The Seven Layers of the OSI Model

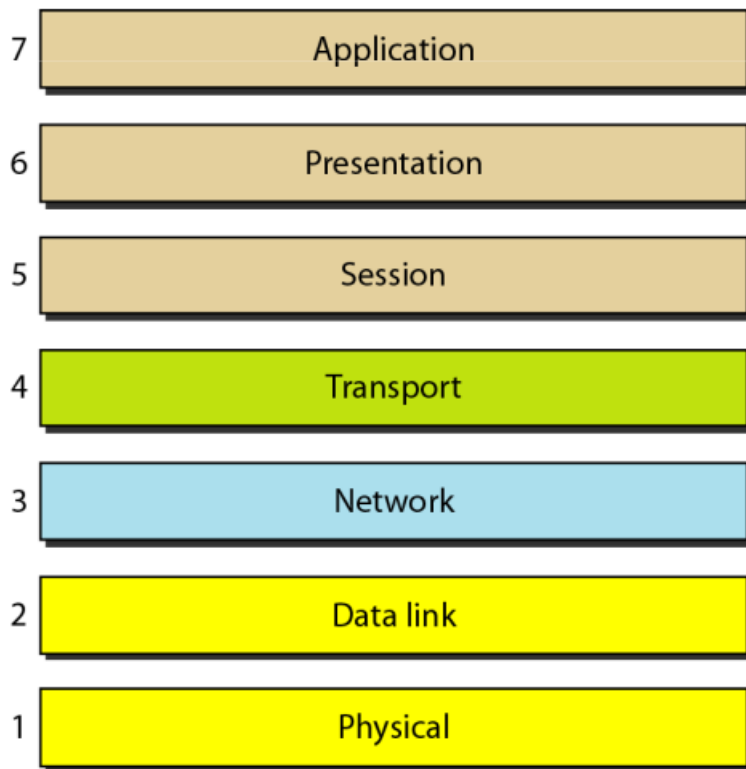
1. **Physical Layer**
2. **Data Link Layer**
3. **Network Layer**
4. **Transport Layer**
5. **Session Layer**
6. **Presentation Layer**
7. **Application Layer**

What is ISO?

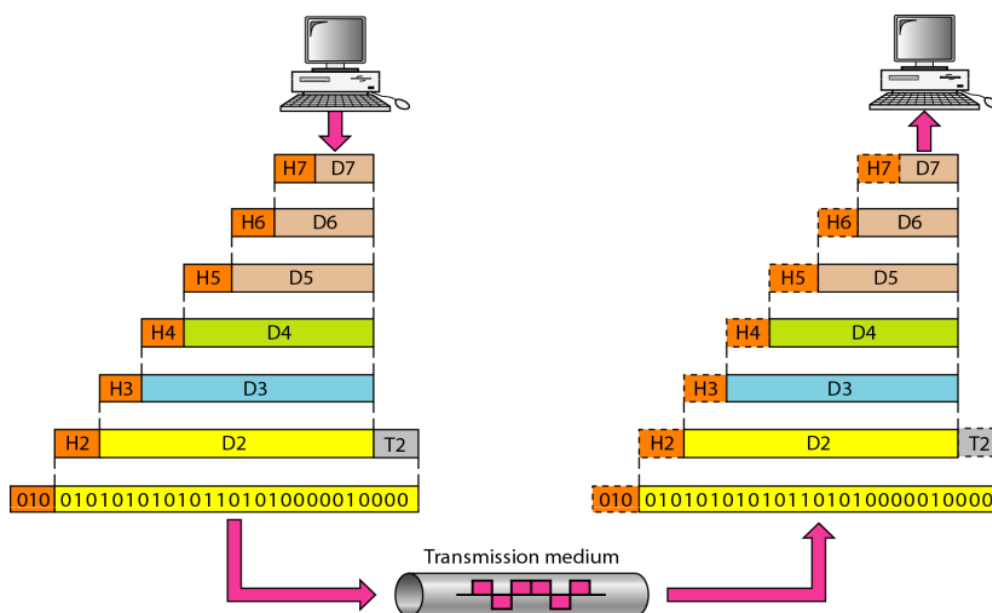
- ISO stands for the International Organization for Standardization.
- It was established in 1947.
- ISO is a group that creates and promotes international standards to help companies and countries work together better.

Layered Tasks in the OSI Model means each layer has its own job to help move data from one place to another. Each layer does its part and then hands the data to the next layer. This way, it's easier to understand, fix, and manage how data travels in a network.

Seven layer of OSI model



An exchange using OSI model

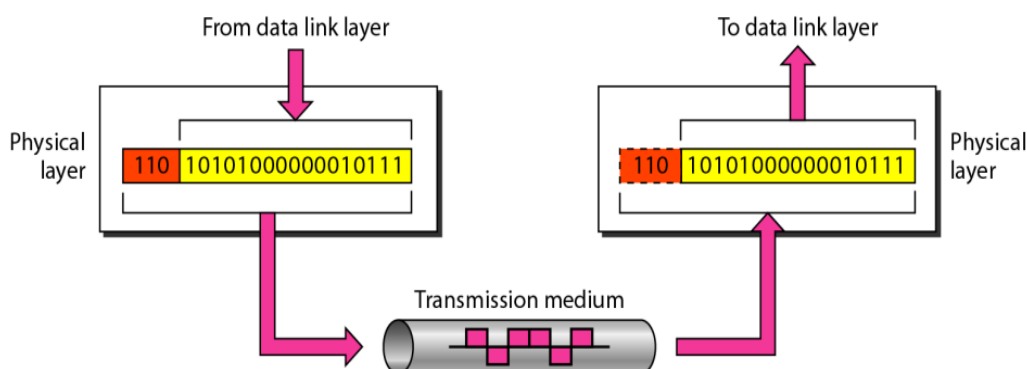


The **OSI Model** (Open Systems Interconnection Model) is a conceptual framework used to understand how different networking protocols interact. It has seven layers, each with a distinct role in handling and processing data across networks. Here's an expanded explanation:

1. Physical Layer

- **What it does:** This is the foundational layer that handles the physical connection between devices. It focuses on transmitting raw data (bits) over physical mediums such as cables or wireless signals. It includes the setup, maintenance, and disconnection of physical links between devices.
- **Example:** Imagine it as the actual cables, connectors, and even light or radio waves that carry electrical signals from one device to another.

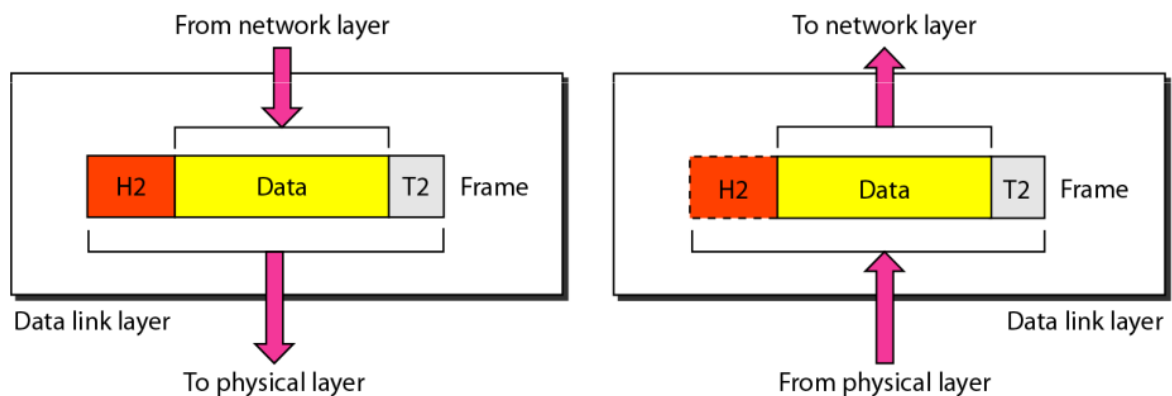
The physical layer is responsible for movements of individual bits from one hop (node) to the next.



2. Data Link Layer

- **What it does:** This layer is responsible for node-to-node data transfer and error correction in a local network. It ensures that data packets are free from errors as they move between devices within the same network. The data link layer also regulates access to the physical medium if multiple devices share it.
- **Example:** Like a local postal service ensuring that a package moves from one address to another within the same city or area, verifying its correct delivery.

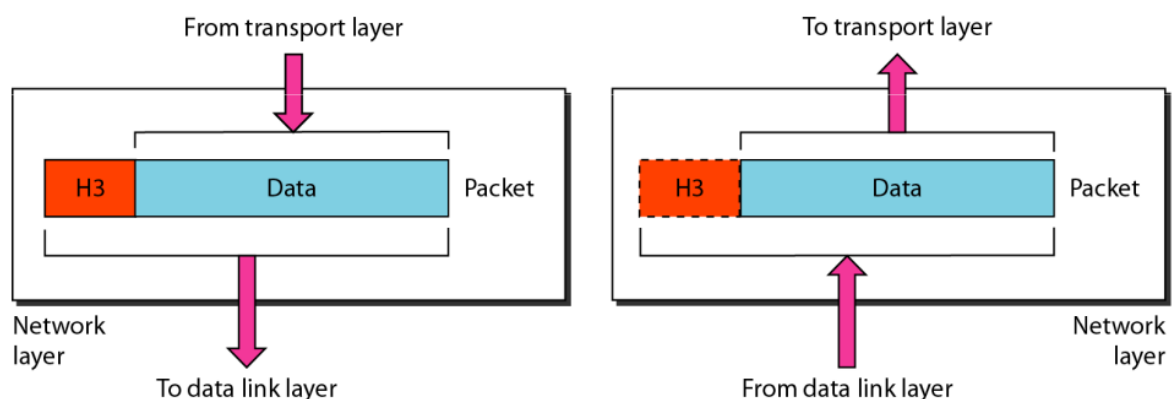
The data link layer is responsible for moving frames from one hop (node) to the next.



3. Network Layer

- **What it does:** The network layer routes data between different networks and decides the best path the data should take to reach its destination. It's like a GPS that selects the optimal route for the data to travel, regardless of the network boundaries.
- **Example:** Similar to GPS directing a car from city to city or country to country, ensuring it finds the most efficient route to reach the final destination.

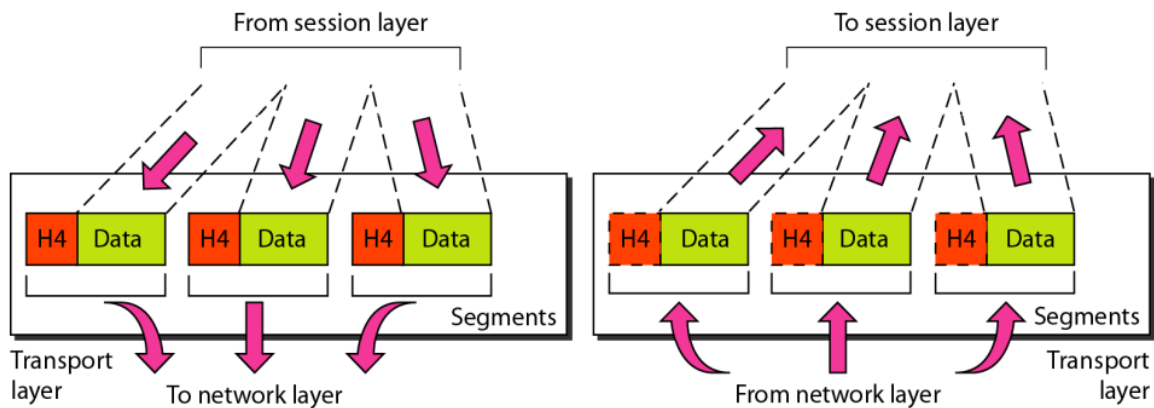
The network layer is responsible for the delivery of individual packets from the source host to the destination host.



Transport Layer

- **What it Does:** The transport layer makes sure that data sent over the internet gets to its destination safely and in the right order. It also controls how fast data is sent to avoid overloading the network. This is very important for things like online banking, where every piece of information matters.
- **Example:** You can think of it like a delivery service that checks to ensure all packages arrive without any getting lost or damaged, and that they are delivered in the correct order.

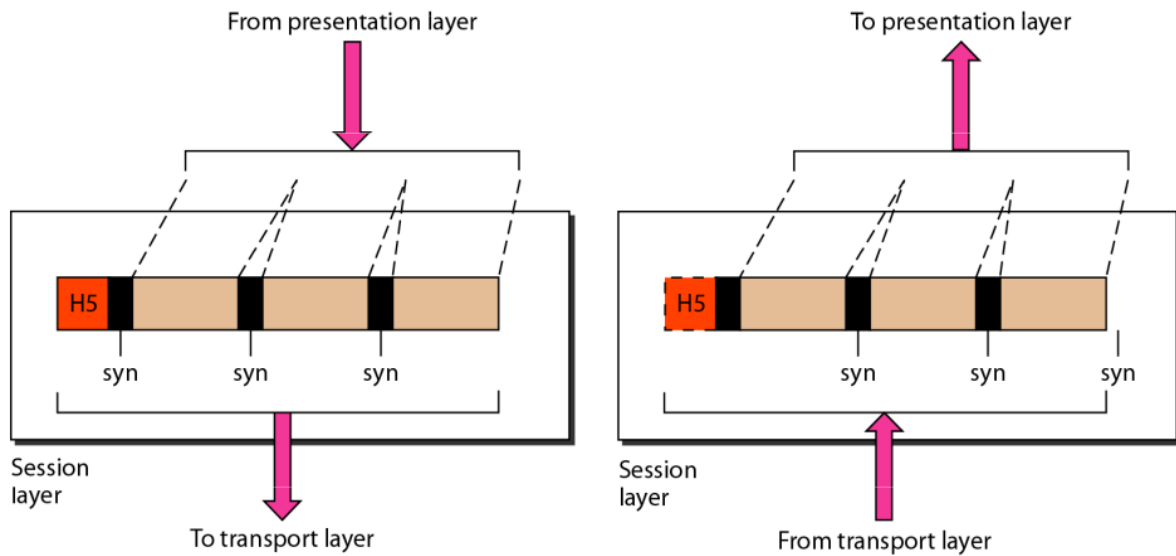
The transport layer is responsible for the delivery of a message from one process to another



5. Session Layer

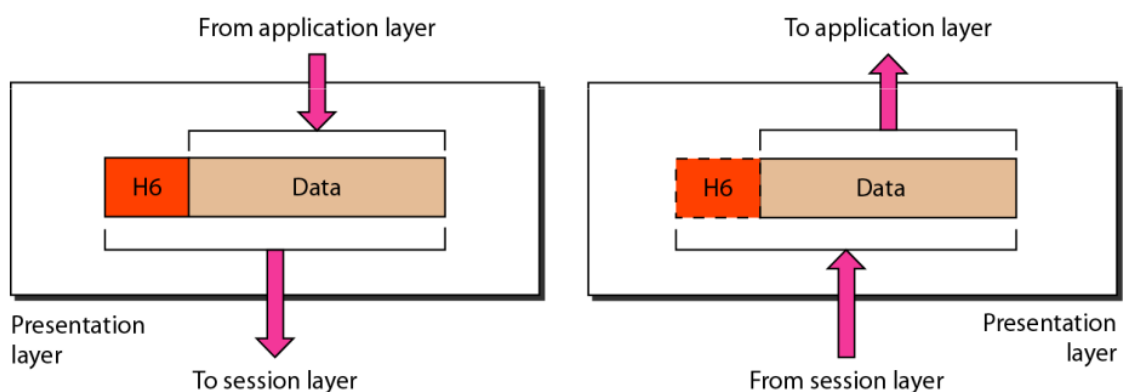
- **What it does:** The session layer handles the communication between two devices (or applications) by setting up, maintaining, and closing sessions. It ensures that data is exchanged smoothly and keeps track of who is talking to whom.
- **Example:** Think of it like a conversation moderator who makes sure both people get a turn to speak and helps manage any interruptions

The session layer is responsible for dialog control and synchronization.



6. Presentation Layer

- **What it Does:** The presentation layer formats and translates data for the application layer. It handles tasks like encryption, compression, and converting data formats so that different systems can understand each other. This is crucial for interpreting data correctly and ensuring security.
- **Example:** Think of it like a translator who makes sure information is conveyed accurately in a language that both parties understand, such as converting text to images or encrypting sensitive data.

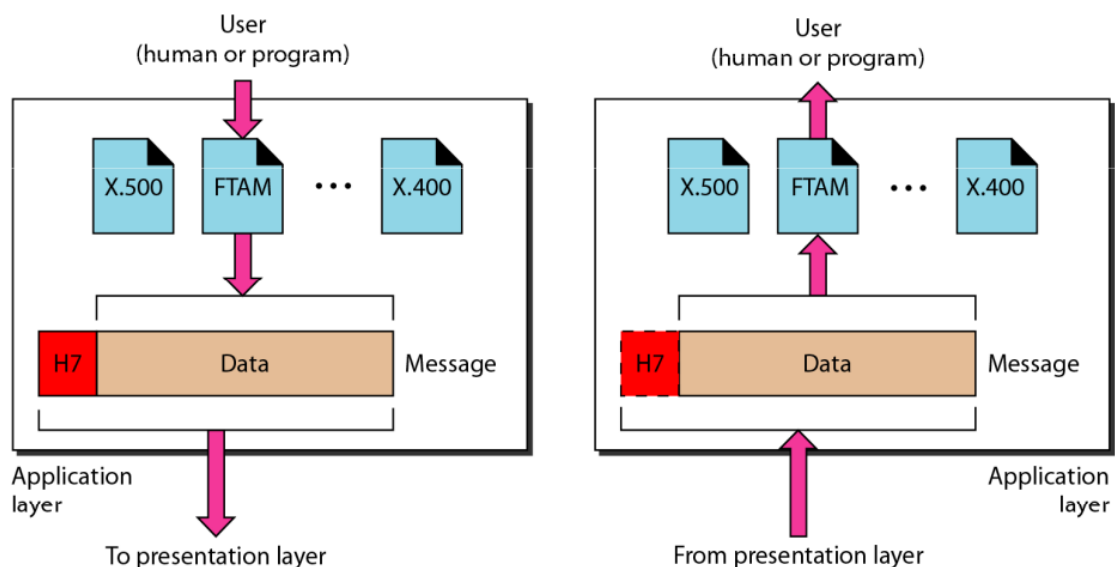


The presentation layer is responsible for translation, compression, and encryption.

7.Application Layer

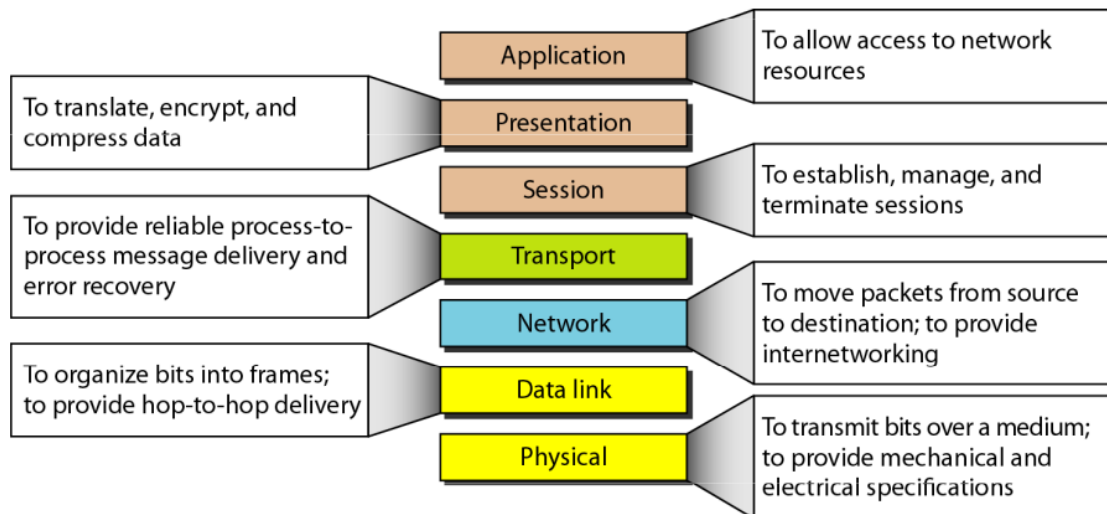
- **What it Does:** The application layer is the closest layer to the end user. It directly interacts with software applications that people use, like web browsers or email clients. This layer identifies communication partners and provides network services to these applications.
- **Example:** Think of it like the apps on your smartphone, where you can easily access services like messaging, browsing, or streaming. It puts all the networking processes right at your fingertips.

The application layer is responsible for providing services to the user.



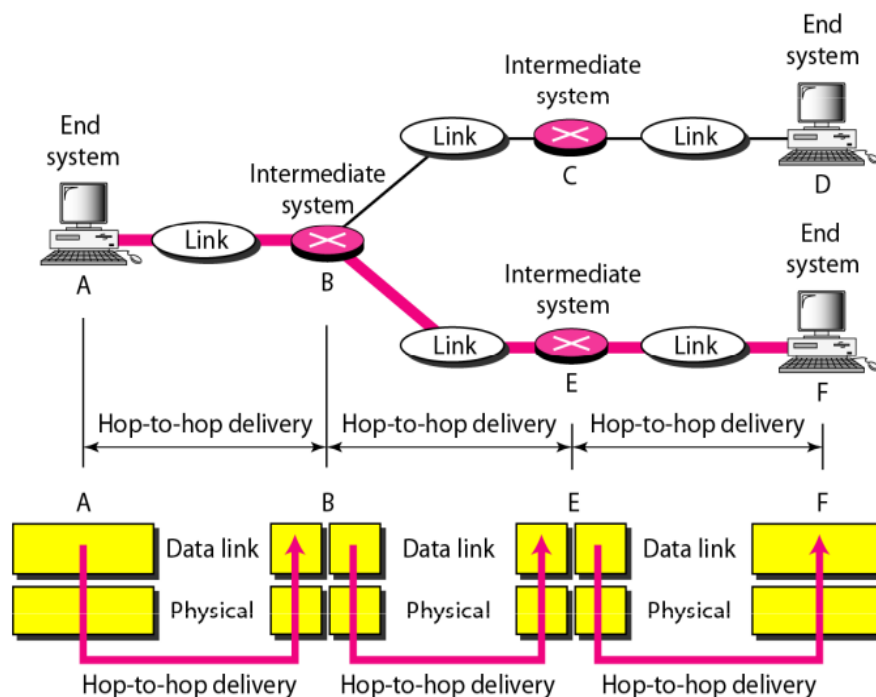
Each layer relies on the one below it, creating a stack where data moves up and down through these layers as it's transmitted or received.

Summary of Layers



Hop-to-hop delivery

Hop-to-hop delivery is a networking concept where data is transmitted from one device to the next intermediate device, rather than being sent directly from the source to the final destination. The data "hops" from one device to the next, using the Physical and Data Link layers to physically transmit the information over the network connections, until it reaches the final destination. This approach allows for more flexible and efficient routing of data through the network, compared to trying to send it directly from start to finish.



The key aspects of the hop-to-hop delivery shown in the image are:

1. **Intermediate systems:** The network comprises multiple intermediate systems (labeled A, B, C, E, and F) that act as relay points for the data transmission.
2. **Links:** The intermediate systems are connected to each other via "Links", represented by the arrows in the diagram.
3. **Data flow:** The data is transmitted from one end system to the other through a series of hop-to-hop deliveries, where the data is passed from one intermediate system to the next.
4. **Physical and Data Link layers:** At each hop, the data is transmitted through the Physical and Data Link layers of the network, as shown by the yellow and gray blocks representing these layers.

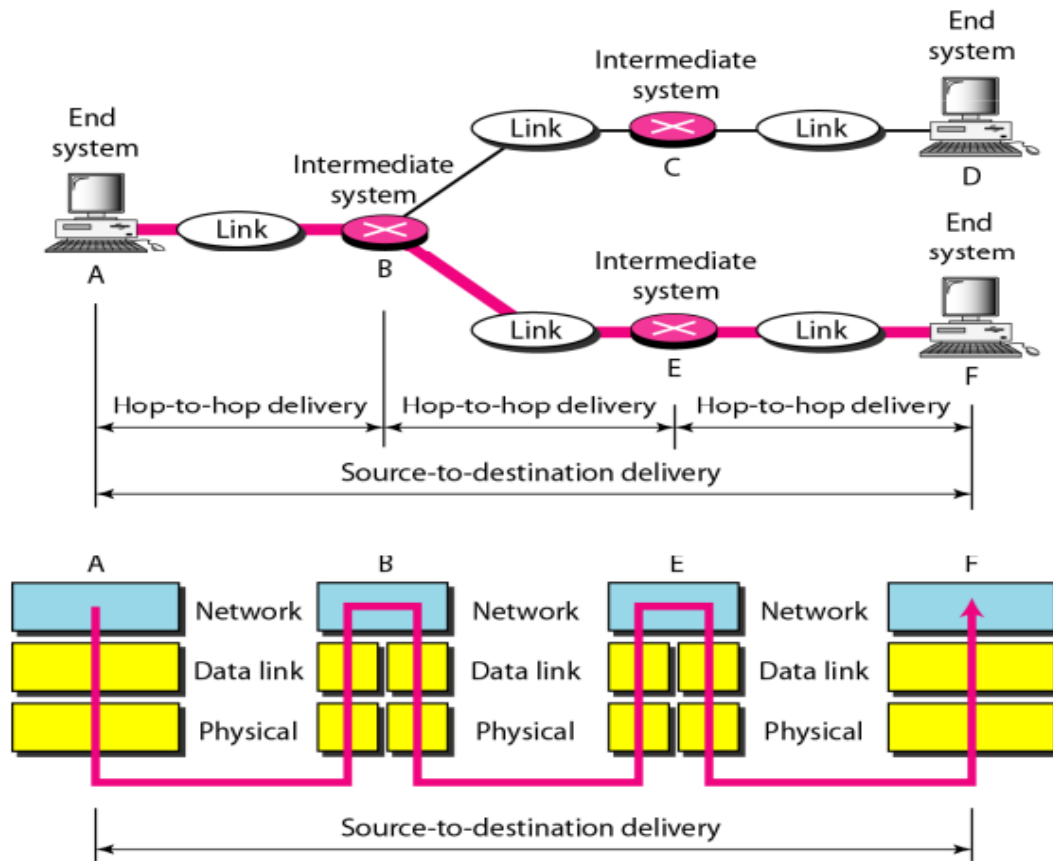
The image demonstrates how the data is transferred between the end systems (A and F) through the intermediate systems (B, C, E) using the hop-to-hop delivery mechanism, with each intermediate system responsible for forwarding the data to the next hop in the network.

Source-to-destination delivery

Source-to-destination delivery means that data is sent directly from the starting device (source) to the final device (destination) without stopping at any other devices along the way. In this method, the source handles all the routing needed to get the data to its destination, making it potentially faster since there are no intermediate stops. However, it requires the source to know the best path through the network. This contrasts with hop-to-hop delivery, where data moves through several devices before reaching its final destination.

Comparison

For large or complex networks, hop-to-hop delivery is generally better due to its flexibility and efficiency. However, for smaller or simpler networks, direct delivery may be faster and easier. The best choice depends on the specific requirements of the network and the type of data being transmitted.



Here are two data delivery methods as follows:

1. Hop-to-hop delivery:

- The data is transmitted from one device to the next, passing through intermediate devices B, C, and E before reaching the final destination F.
- Each intermediate device is responsible for forwarding the data to the next "hop" in the network.

2. Source-to-destination delivery:

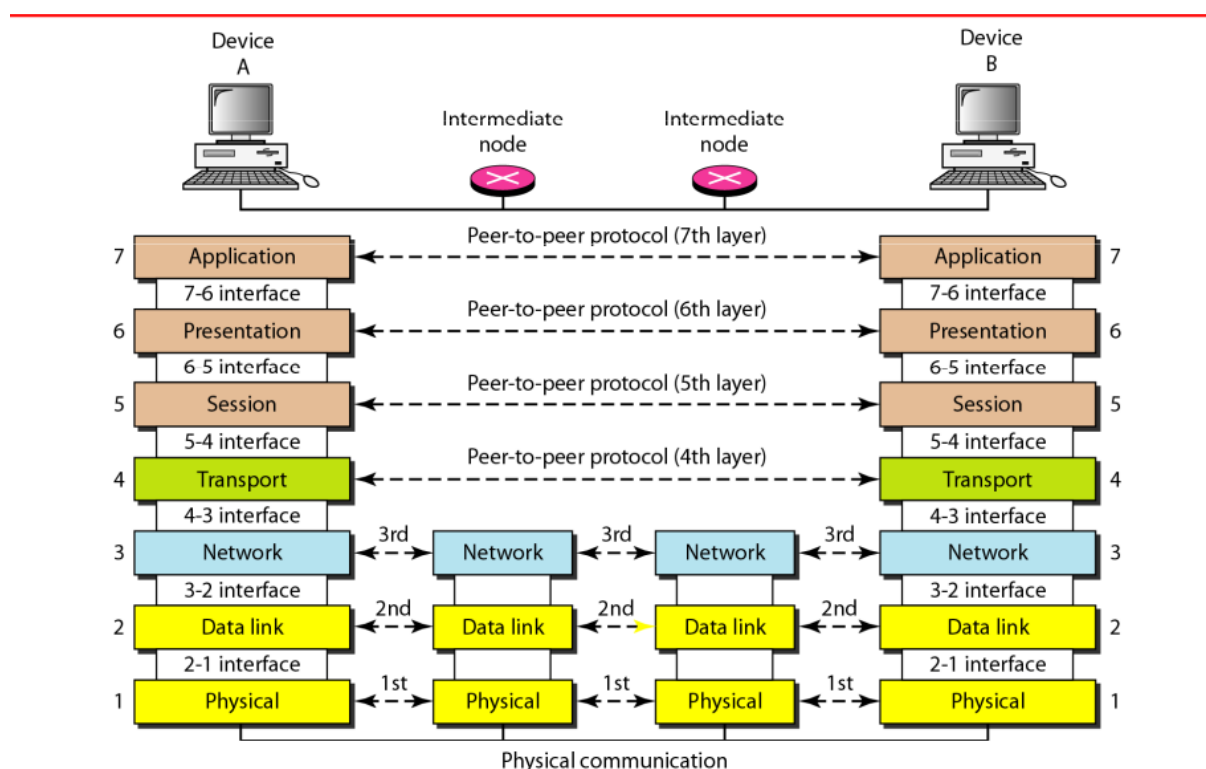
- The data is sent directly from the starting device A to the final destination F, without going through any intermediate devices.
- The source device A handles all the routing and delivery to reach the final destination.

The choice between the two approaches depends on factors like the size and complexity of the network.

Why is the OSI Model Important?

- **Standardization:** Helps different types of devices and software communicate with each other.
- **Troubleshooting:** Makes it easier to find and fix problems by identifying which layer is causing issues.
- **Compatibility:** Ensures that different technologies can work together smoothly.

The interaction between layers in the OSI model



The key aspects illustrated in the image are:

1. **Peer-to-peer protocols:** The communication between the corresponding layers in the two devices is facilitated by peer-to-peer protocols, shown by the horizontal arrows between the layers.
2. **Interfaces:** Each layer has a defined interface that allows it to interact with the layer above and below it. These interfaces are shown by the horizontal lines between the layers.

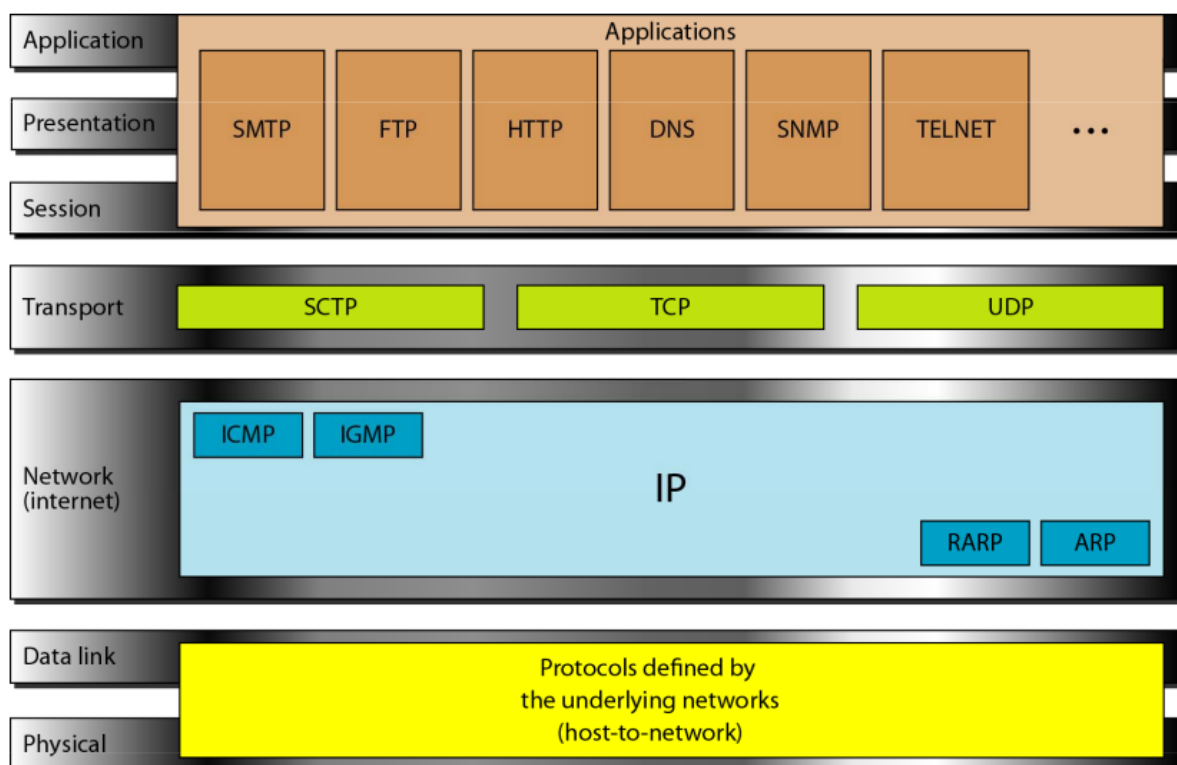
3. **Physical communication:** The bottom-most Physical layer represents the physical communication channel between the two devices.
4. **Network components:** The 3rd layer, Network, and the 2nd layer, Data link, are shown to have additional network components (represented by the "3rd" and "2nd" labels) that facilitate the communication at those levels.

The TCP/IP Protocol Suite

The TCP/IP Protocol Suite is a set of communication protocols used to connect devices on the internet and in other networks. Named after its two main protocols, TCP (Transmission Control Protocol) and IP (Internet Protocol), it provides a foundation for data exchange over interconnected networks.

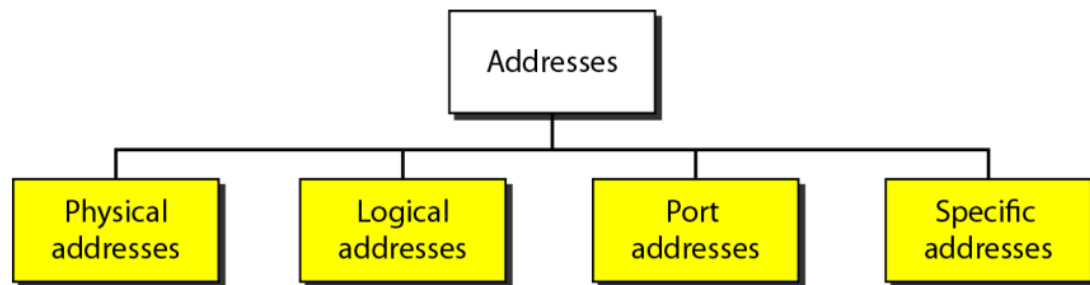
The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made up of five layers: physical, data link, network, transport, and application.

TCP/IP and OSI model

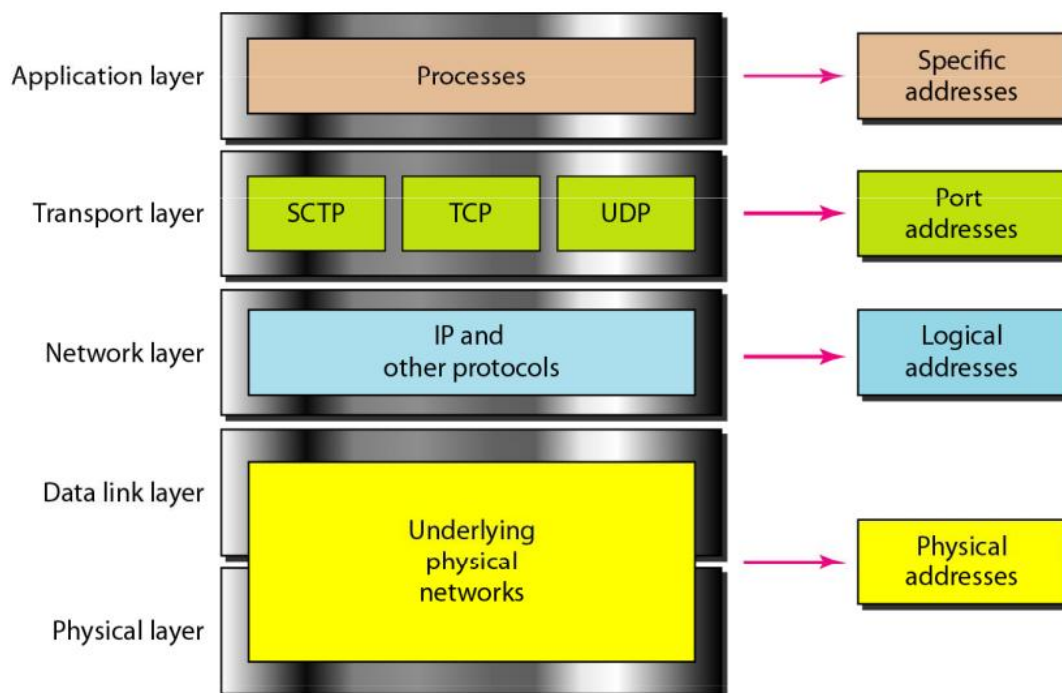


Addresses in TCP/IP

Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.

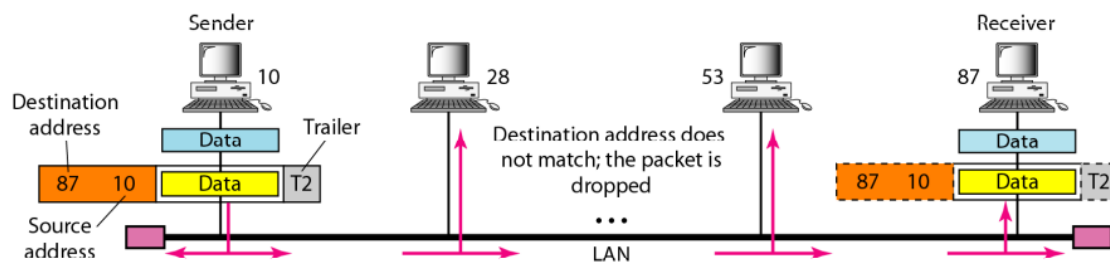


Relationship of layers and addresses in TCP/IP



Physical Address

A physical address refers to a unique identifier assigned to a device on a network, typically in the form of a MAC (Media Access Control) address. This address is used at the Data Link layer of the OSI model and allows devices to communicate within a local network.



In Figure , a node with the physical address 10 sends a frame to a node with the physical address 87. The two nodes are connected by a link in a bus topology LAN. As the figure shows, the computer with the physical address 10 is the sender, and the computer with the physical address 87 is the receiver.

Format: A MAC address is typically represented as a 48-bit (6-byte) physical address, consisting of 12 hexadecimal digits grouped into six pairs (e.g., 00:1A:2B:3C:4D:5E), with each byte (2 digits) separated by a colon.

IP Address

An IP address (Internet Protocol address) is a unique identifier assigned to each device connected to a network that uses the Internet Protocol for communication. It serves two main functions: identifying the host or network interface and providing the location of the device in the network.

Format:

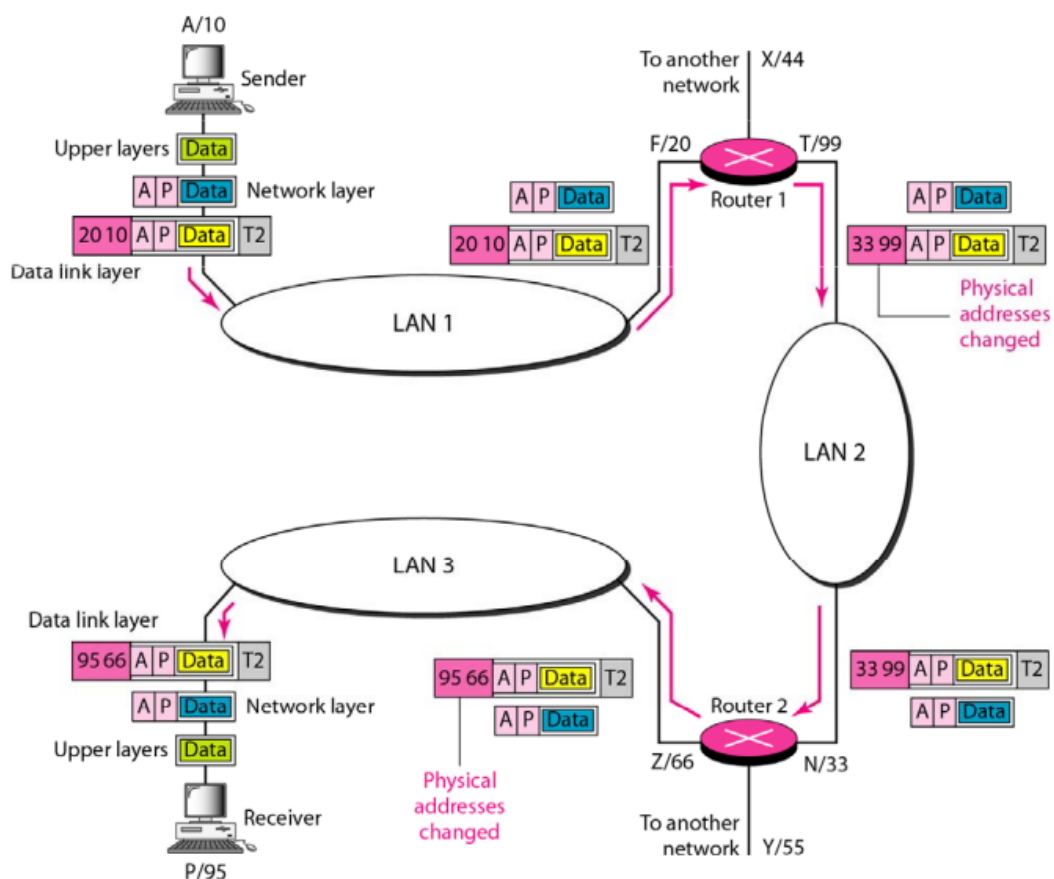
- **IPv4:** Typically written as four decimal numbers (e.g., 192.168.1.1), each ranging from 0 to 255.
- **IPv6:** Written as eight groups of four hexadecimal digits (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

Types:

- **Public IP Address:** Assigned to devices directly connected to the internet and can be accessed globally.
- **Private IP Address:** Used within local networks and not routable on the internet (e.g., 192.168.x.x).

Function: IP addresses enable devices to send and receive data over the internet, allowing them to locate each other and establish connections.

Figure shows a part of an internet with two routers connecting three LANs. Each device (computer or router) has a pair of addresses (logical and physical) for each connection. In this case, each computer is connected to only one link and therefore has only one pair of addresses. However, each router is connected to three networks (only two are shown in the figure), so each router has three pairs of addresses, one for each connection



Port Addresses

A **port address** is a number that identifies specific services or applications on a device connected to a network. It works with an IP address to ensure data reaches the correct application.

A 16-bit port address represented as one single number.Ex:753.

Function: Port addresses route data to the correct application, enabling effective communication.In short, port addresses are crucial for directing network traffic to the right services on devices.

Note: The physical addresses will change from hop to hop, but the logical addresses usually remain the same.

Figure shows two computers communicating via the Internet. The sending computer is running three processes at this time with port addresses a, b, and c. The receiving computer is running two processes with port addresses j and k. Process a in the sending computer needs to communicate with process j in the receiving computer. Note that while physical addresses change from hop to hop, logical and port addresses remain the same from the source to the destination.

